Split Protocol or Feature Transform? Explaining Generalisation Gaps in Acoustic Honeybee Colony Classification

Atharve Dahima[a,1], Srajan Mishra[a,1]

[a] *Independent, India*

Abstract

Acoustic classifiers for honeybee colony state are frequently reported with accuracy above 90%, yet the same models degrade when whole hives are held out. Two explanations compete: split leakage, where near-duplicate rows cross the train-test boundary, and feature encoding, where microphone gain acts as a hive fingerprint.

We test both on the MSPB dataset. A reproduction of the published protocol uses 210,600 telemetry rows from 306 inspection windows. A random row split reaches 86.8 percent accuracy. Removing duplicate rows, grouping the split by hive, or aggregating to the inspection level each removes most of that gain.

We then cross four split rules with three feature transforms at inspection granularity, holding data, label, models and metric fixed. The most permissive split scores 0.7444 macro F1, and the strict hive holdout scores 0.7445. The pre-registered inflation claim is falsified ($p = 0.073$ over 15 model-transform pairs). Absolute decibel features do not leak either. The

[1]Corresponding author.

only large effect is indoor environment, worth 8.5 to 11.3 macro-F1 points, and it transfers to an unseen apiary.

Three of the eight canonical acoustic bands receive no data, because the sensor covers 122 to 580 Hz.

At row level the leak returns, and it is capacity-dependent. A random forest gains +9.2 macro-F1 points from a split that it loses under hive hold-out, and logistic regression gains +1.1. Aggregating before splitting removes the leak and raises the honest score. Report the unit of analysis with the score.
Keywords: honeybee, acoustic monitoring, data leakage, cross-validation, group holdout, reproducibility

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1. Introduction

Pollinator decline threatens crop production and wild plant reproduction [1, 2]. Continuous hive monitoring lets a beekeeper observe colony state without opening the hive [3].

Machine learning models that classify honeybee colony state from hive audio are frequently reported with accuracy above 90% [4, 5, 6, 7]. When the evaluation holds out whole hives, the same task becomes much harder [8]. The gap between the two protocols is large enough to change the conclusion of a study.

Two explanations are available, and the literature rarely separates them.

The first is *split leakage*. Adjacent recordings from one hive are near duplicates. A random row split places some of them in training and others in test. The model then recognises the hive rather than the colony state.

This failure mode is well documented outside apiculture [9, 10, 11]. Model selection adds a related bias when the same data choose the model and estimate its score [12, 13].

The second is *feature encoding*. Microphone placement, cavity resonance and sensor gain give each hive a stable spectral fingerprint. Features expressed in absolute decibels retain that fingerprint. A model can therefore learn hive identity from the feature vector even when the split itself is clean. Gain-invariant features, such as band power relative to the row mean, remove the fingerprint.

This paper asks which explanation dominates, and answers it by experiment rather than by argument. We hold the data, the label, the models and the metric fixed. We vary two things independently: the split rule, and the gain invariance of the feature transform. No previous study in this domain varies both in one design.

Our contributions are:

1. A direct reproduction of the published row-level protocol on this dataset, followed by removal of its sources of optimism one at a time.
2. A four-rung split ladder that isolates hive leakage from apiary-level environment shift.
3. A two-way comparison of gain-invariant and gain-retaining features across every rung.
4. A full accounting of a public dataset's usable acoustic range, including the three canonical bands that no sensor in the dataset covers.
5. A pre-registered analysis, with the falsification outcome reported as obtained.

1.1. Related work

Acoustic hive monitoring has a long record of high reported accuracy. Early benchmarks compared classical spectral features with shallow classifiers for hive sound recognition [4]. Deep networks followed, first for swarm activity from IoT microphones [5], and later for colony classification. One recent study reports 98.31 % accuracy on data described as unseen [6]. A survey of embedded beekeeping systems collects these results and notes that evaluation protocols differ between studies [7]. Recent work keeps the time dimension of the modulation spectrum and reports better cross-hive transfer than earlier handcrafted features [14]. Precision beekeeping sets out its own engineering constraints [15].

The protocol question is not new to this domain. The Bee Together study joined several public bee audio datasets to test extrapolation to hives that a model has not heard [8]. That work establishes cross-hive transfer as the hard setting. It does not separate the mechanisms that produce the gap.

Outside apiculture the mechanisms are well characterised. Leakage through near-duplicate records is formalised by [9] and surveyed at scale by [10]. Grouped cross-validation for dependent observations is set out by [11].

Sensor fusion is the other established direction. Indoor environmental sensing has been used for colony monitoring [16] and for queen detection at the edge [17]. In-hive thermal patterns have been used to detect colony events [18]. Vibrational analysis has been used for long-term health assessment [19, 20]. We are not aware of a study in this domain that varies the split rule and the feature transform independently.

2. Data

We use the MSPB dataset [21], released under doi:10.1038/s41597-024-03695-1 and archived at Zenodo record 11398835.

2.1. Telemetry

The dataset contains 318 labelled inspection records derived from 1,836,914 raw telemetry rows. Two files partition the year: D1 covers 16 April to 5 November 2020 and holds 960,809 rows. D2 covers 6 November 2020 to 14 April 2021 and holds 876,105 rows.

The nominal sampling interval is five minutes in the dataset documentation. The observed median interval is 15 minutes, measured across both files. We report the observed value throughout.

Each row carries 16 acoustic bins spanning 122.07 Hz to 579.83 Hz with a uniform spacing of 30.52 Hz. Values are already logarithmic, in decibels, and many are negative. No additional logarithm is valid.

2.2. Phenotypic label

Colony strength is measured in six inspection rounds between 9 June and 21 August 2020. Each inspection records the number of frames covered by bees across all chambers. We follow the dataset authors and label a colony *strong* when the total frame count is at least 20.0.

The raw telemetry covers 94 sensor tags. Only 53 hives carry a colony-strength label. All results in this paper describe those 53 hives, and the two counts are not interchangeable.

2.3. Feature construction

Each inspection is represented by one row. Features are computed from a seven-day window that ends strictly before the inspection timestamp. Inspections are a median of 14.0 days apart, so the windows do not overlap and a row can never see its own future.

The 16 source bins are grouped into eight canonical bands spanning 30 Hz to 2000 Hz. This mapping has a consequence that matters for interpretation: the MSPB microphone covers 122 to 580 Hz, so bands 1 to 5 receive data and bands 0, 6 and 7 receive none. Three of the eight bands are therefore synthetic, held at a numerical floor. We report results with all eight bands, as the original pipeline does, and we state the occupancy in Section 5.4.

A window must contain at least ten telemetry samples to be retained. 12 of 318 inspections failed this test, leaving 306 rows across 53 hives. The label balance is 194 strong and 112 weak.

2.4. Apiaries

The hives sit at two apiaries, `nectar-bh121` with 27 hives and `nectar-bh131` with 26 hives. A hive never appears at both sites. We verified this after mapping sensor tags to physical hive identifiers, because raw tag numbers do overlap between the two files. The overlap is an artefact of per-site tag numbering and does not survive the lookup table.

3. Methods

3.1. Design

The experiment crosses four split rules with three feature transforms, giving twelve conditions. Data, label, models, seeds and metric are identical

in every condition. Only the split and the transform change.

3.2. Split ladder

The four rungs are ordered from most to least permissive.

P0, random rows. Stratified five-fold split over rows, ignoring hive membership. A hive appears in training and test with high probability. This rung reproduces the protocol that produces the high scores in the literature.

P1, temporal holdout. Training uses the first five inspection rounds and test uses the last. Hives are shared, but no test row precedes a training row. This rung isolates temporal generalisation.

P2, unseen hive. Leave-one-hive-out cross-validation. Each fold trains on all hives but one. No hive contributes to both sides.

P3, unseen apiary. Train on one apiary and test on the other, in both directions. Each row is predicted exactly once. This rung adds environment shift to hive shift.

The decomposition follows directly: the P1 to P2 difference measures hive leakage, and the P2 to P3 difference measures environment leakage.

3.3. Feature transforms

Three transforms share the same seven-day window.

Relative bands. Eight band powers, converted to decibels, minus the row mean across bands. This operation removes broadband gain and therefore removes the microphone fingerprint. It yields 16 features: the mean and standard deviation of each band over the window.

Absolute bins. The 16 source bins in decibels, averaged and summarised over the window. This transform retains gain. It yields 32 features.

Relative bands with environment. The 16 band features plus six indoor measurements: temperature mean, standard deviation, minimum and maximum, and humidity mean and standard deviation. This transform tests whether colony-internal environment adds information that survives a change of apiary.

3.4. Models

Five classifiers span the range used in this literature: logistic regression, random forest with 300 trees, histogram gradient boosting, a two-hidden-layer perceptron with 32 and 16 units, and a radial-basis-function support vector machine. Scaled models receive a standard scaler fitted inside the pipeline, so scaling never sees test data. Class imbalance is handled by class weighting only, never by resampling.

3.5. Estimation and uncertainty

Reported values are pooled out-of-fold predictions, not means of per-fold scores. Per-fold means overweight small folds, and the hive holdout produces 53 folds of roughly six rows each.

Uncertainty is a percentile bootstrap over hives with 2000 resamples [22]. Resampling rows would understate the interval, because rows from one hive are dependent, and the variance of a cross-validation estimate is itself uncertain [23]. Comparisons across models use a two-sided Wilcoxon signed-rank test [24, 25].

3.6. Pre-registration

The analysis plan, the predictions and the decision rules were written to disk before any model was fitted. The plan fixed the headline model as the best performer at P2, chosen before P3 was inspected, to prevent selecting the model that transfers best. The separation of accuracy estimation from model selection is an old concern [26].

3.7. Reproduction of the published protocol

The ladder aggregates before it splits, so it cannot show what a row-level protocol does. We therefore reproduced the published protocol directly.

Every telemetry row in a retained window enters, with no cap. Features are the sixteen absolute decibel bins, which are the closest available analogue of the spectral inputs that the literature reports. Four protocols share the same features and the same models.

Protocol A splits rows at random. Protocol A2 keeps the random split and retains one row per window. Protocol B keeps every row and groups the split by hive. Protocol C aggregates to one row per inspection and groups by hive.

Accuracy is the reported metric, because accuracy is what the high-scoring studies report. Macro F1 and AUROC are recorded in the result file for completeness.

3.8. Row-level control

The ladder uses one row per inspection. A second experiment keeps the same windows, the same label and the same gain-invariant transform, and

changes only the unit of analysis. Each window contributes at most 300 telemetry records.

It compares a random row split with a five-fold hive-grouped split. Three models are used, because the perceptron and the support vector machine do not scale to this row count.

4. Results

4.1. The split rule does not change the score

Table 1 reports macro F1 for the model that scored best at P2, selected before P3 was inspected.

The random row split, which places the same hive in training and test, scores 0.7444. The unseen-hive split scores 0.7445. The difference is -0.0002, and the confidence intervals overlap almost completely.

The pattern holds for every model and every transform. Across 15 model-transform pairs the mean P0 to P2 gap is +0.0052 and the standard deviation is 0.0108. The largest absolute gap is 0.0259, which is smaller than the narrowest bootstrap interval for any rung in Figure 1 (0.0921). 5 of the 15 gaps are negative, so the direction is not uniform. A two-sided Wilcoxon signed-rank test gives $p = 0.073$, which does not reject a median gap of zero at the five percent level.

The pre-registered rule required a gap of twice the bootstrap standard error. The observed mean gap is +0.0052, a small fraction of that threshold. Hive leakage does not inflate the score when the unit of analysis is an inspection.

Table 1: Macro F1 by split rung and feature transform, for the headline model (HistGradientBoosting). The P0 rung leaks hives and the P2 rung does not. Bootstrap intervals for each rung appear in Figure 1.

Transform	P0 random	P1 temporal	P2 unseen hive	P3 unseen apiary
Relative bands	0.7444	0.3892	0.7445	0.7258
Absolute bins	0.7505	0.3287	0.7245	0.6954
Bands + environment	0.8445	0.4074	0.8292	0.7856

4.2. Gain invariance is not the mechanism either

If microphone gain were the leaked signal, absolute decibel features would score higher under a random split and lower under hive holdout. They do not. Absolute bins score 0.7505 at P0 and 0.7245 at P2, a difference of +0.0259.

Removing the row mean therefore does not remove a leak. It removes a nuisance parameter, and the effect on score is close to zero for tree models. Logistic regression benefits from absolute bins by +4.5 points at P0 and +5.3 at P2, and by +1.3 at P3, which suggests the relative transform discards information that a linear model can use. That is a modelling choice, not a leakage effect.

4.3. Environment features are the only large effect

Adding six indoor measurements raises macro F1 from 0.7445 to 0.8292 at P2, an increase of +8.5 points for the headline model. Across the five models the gain at P2 runs from 8.5 to 11.3 points. No split manipulation in this study produces an effect of that size.

The gain survives the apiary change. At P3 the environment-augmented

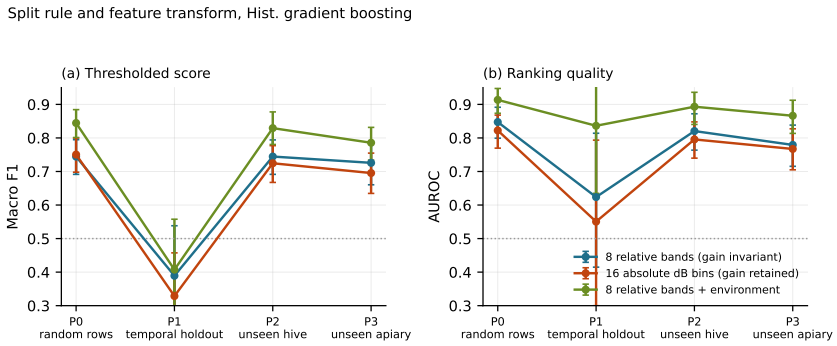


Figure 1: Macro F1 and AUROC across the four rungs for the headline model. Error bars are percentile bootstrap intervals over hives. Panel (a) shows the thresholded score; panel (b) shows ranking quality. The random-row rung and the unseen-hive rung are indistinguishable in both, which means hive leakage is not present at this unit of analysis. Only the temporal rung degrades, and it degrades on both measures for most models.

model still scores 0.7856, and its area under the receiver operating characteristic curve reaches 0.8662.

This is a positive result about colony physiology. Brood-nest temperature and humidity are colony-level properties, and they carry information that transfers to a site the model has never seen.

4.4. The temporal holdout is the only rung that degrades

P1 is the exception to the flat pattern, and it is the least reliable rung. Macro F1 falls to 0.3892 for the headline model, with a bootstrap interval from 0.2448 to 0.5383 on 48 test rows.

Ranking also degrades for 4 of the 5 models, reaching between 0.6237 and 0.6516. That is a genuine generalisation failure, not a threshold artefact.

Logistic regression is the exception. It holds AUROC at 0.8049 while its macro F1 falls to 0.3516. A model that ranks well but classifies badly at a

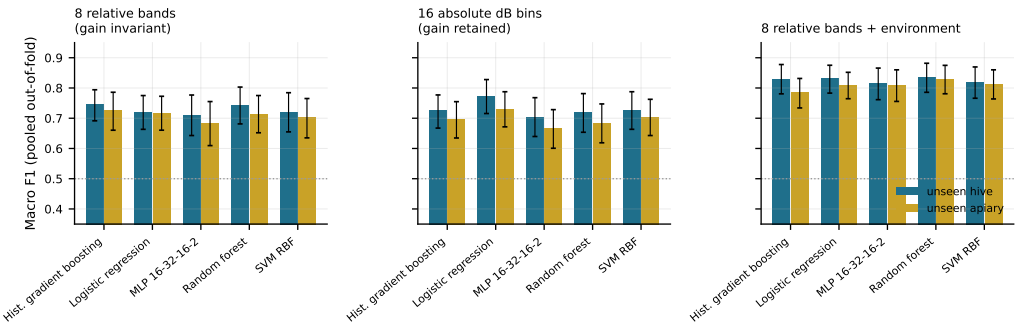


Figure 2: Macro F1 at the two strict rungs for every model, split by transform. The transform changes the score more than the rung does.

fixed cut-off is miscalibrated rather than uninformed.

With 48 test rows we report both patterns and interpret neither strongly. The correct summary is that the last inspection round is the hardest, and that the difficulty can appear either as lost ranking or as a shifted operating point, depending on the model.

4.5. Three of eight bands carry no data

Figure 3 shows the occupancy of the eight canonical bands. The MSPB microphone spans 122 to 580 Hz. Bands 0, 6 and 7 fall outside that range and receive no bins. They are held at a numerical floor and contribute no measurement.

Any study that reports an eight-band biological decomposition on this dataset is reporting five measured bands and three constants.

4.6. Restoring the row-level protocol brings the leak back

The null result above has an obvious alternative explanation: the inspection-level dataset may be too small and too smooth to leak. We tested that by

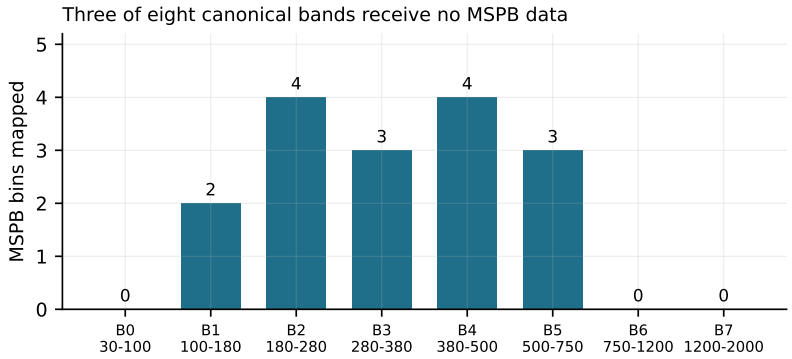


Figure 3: Number of MSPB bins mapped to each canonical band. Bands 0, 6 and 7 receive no data because the sensor range is 122 to 580 Hz.

returning the unit of analysis to the individual telemetry record.

Each of the 306 windows contributes at most 300 records, giving 89372 rows across 53 hives. The transform, the label and the models are unchanged, apart from dropping the two models that do not scale to this row count. Only the split differs.

Leakage reappears, and it is model-dependent. The random forest gains +9.2 macro-F1 points from a random row split that it loses under a hive holdout (0.7754 to 0.6832). Histogram gradient boosting gains +5.5 points. Logistic regression gains +1.1.

The ordering is the useful part. A high-capacity model can memorise near-duplicate rows; a linear model cannot. This reproduces the mechanism that the leakage literature describes, and it shows that the mechanism requires both the row-level protocol and a model with enough capacity to exploit it.

Two conclusions follow, and they point in the same direction. First,

aggregation to the inspection window removes the leak for every model tested. Second, aggregation also raises the honest score: the inspection-level hive holdout reaches 0.7445, while the row-level hive holdout reaches only 0.6852.

Aggregating before splitting is therefore not merely a defensive choice. It is the better model.

4.7. The published protocol, reproduced and then broken

The row-level control above already used a conservative version of the published protocol: 300 rows per window and gain-invariant features. We therefore reproduced the protocol in its strongest form. Every telemetry row enters, all sixteen absolute decibel bins are used, and the split is random.

The data are 210,600 rows from 306 inspection windows across 53 hives.

Table 2 reports accuracy, because accuracy is the metric that the high-scoring literature reports. Under protocol A the best model reaches 86.8 percent. That value sits in the range reported for hive audio, and it comes from a protocol that places near-duplicate rows on both sides of the boundary.

We then change one thing at a time.

Protocol A2 keeps the random split and removes the duplicate rows, keeping one row per window. Protocol B keeps every row and groups the split by hive. Protocol C aggregates to one row per inspection and groups by hive.

The drop from A to A2 is the duplicate-row effect with the split held fixed. The drop from A to B is the hive-membership effect with the features held fixed. Both are large, and they are not the same thing.

Protocol C is the honest number, and it is the one a deployment should expect.

Table 2: Accuracy under four protocols on identical features. A is the published protocol. A2 removes duplicate rows. B removes hive membership from the test set. C aggregates to one row per inspection. Every model uses the same sixteen absolute decibel bins.

Model	A published	A2 deduplicated	B grouped	C aggregated
Logistic regression	70.4	69.3	69.0	77.5
Random forest	86.8	70.3	73.3	75.2
Extra trees	86.2	69.3	72.8	75.2
Hist. gradient boosting	77.1	66.0	70.7	70.3

4.8. Robustness to the label cut-off

The 20-frame threshold is inherited from the dataset authors. A reviewer can reasonably ask whether the null result depends on it. We repeated the two extreme rungs at three thresholds, 15, 20 and 25 frames, across three transforms and five models. That gives 45 gaps.

The gap ranges from -0.0816 to +0.0858. No cut-off produces an inflation of the size that the row-level protocol produces in Table 2. The null at inspection granularity is therefore not an artefact of the label definition.

5. Discussion

5.1. What the null result means

The pre-registered prediction was that a random row split would inflate macro F1 by at least twice the bootstrap standard error. No model-transform pair reached that threshold. Under the decision rules fixed before the analysis, the claim is falsified at the inspection level.

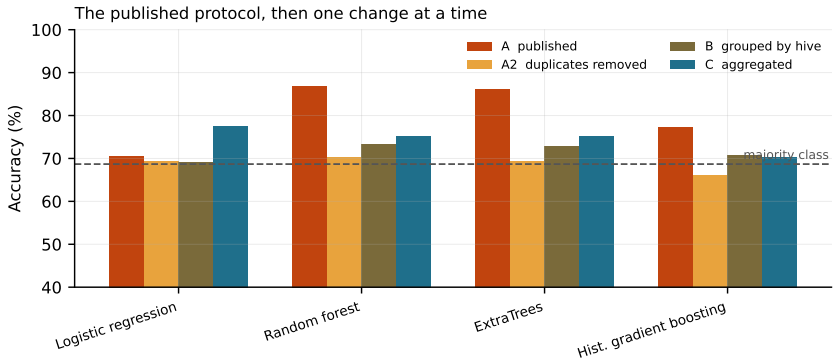


Figure 4: Accuracy under the published protocol and under three repairs. Each repair changes one thing. Removing duplicate rows and grouping by hive each remove much of the gain. Aggregation gives the honest number, and it still clears the majority-class line.

We report the failure rather than reframe it. The result has a clear interpretation: an inspection-level dataset holds at most 6 rows per hive, and those rows are 14.0 days apart. There is nothing for a random split to memorise. The leakage that the literature describes lives in the protocol that splits individual recordings, where thousands of near-duplicate rows belong to one hive. The row-level experiment in Section 4.6 confirms that it reappears there for high-capacity models.

This finding is useful to practitioners. It says that a carefully aggregated feature set is robust to the split, and that the 74.5 to 82.9 percent range measured here for colony-strength prediction is plausibly real rather than an artefact.

5.2. Where the signal actually is

The largest effect in the study is not a leakage effect at all. It is the contribution of indoor temperature and humidity, worth 8.5 to 11.3 macro-

F1 points at P2 and stable across apiaries. Acoustic bands alone reach 0.7445 at P2. The combination reaches 0.8292.

For a system designer this is a concrete instruction: spend the sensor budget on the brood-nest climate before spending it on a wider acoustic band.

5.3. Implications for reported accuracies

Scores above 90 percent in this literature are not automatically wrong, but they require a protocol that makes them possible. A study that classifies individual recordings and splits them at random can reach such numbers without learning anything about colony state. A study that aggregates to the inspection level cannot, and should not be expected to.

The two numbers are not comparable, and a paper that reports one should state which protocol produced it.

We measured the difference directly in Section 4.6. The same sixteen features reach 86.8 percent under a random row split and 73.3 percent when the split respects hive membership. The features did not change. Only the protocol did.

5.4. Limitations

The label is a threshold on frame count, and the threshold at 20 frames is a choice inherited from the dataset authors. A different cut-off would change the base rate and possibly the ranking of models. Section 4.8 shows that it does not change the leakage null.

Colony strength is not constant within a hive. Of the 53 hives, 46 change class at least once across the 6 rounds, which is 86.8 percent. The median

within-hive range of total frames is 14.0 on a 0 to 34 scale. A hive hold-out therefore tests whether within-hive change generalises to new hives, not merely whether hives can be separated. That is the correct test for deployment.

The reproduction in Section 4.7 applies the published protocol to this dataset, not to the dataset of any single prior study. It shows that the protocol can produce high accuracy here. It does not reproduce a specific published number.

Only 53 hives carry labels, from two apiaries in one region and one season. The apiary holdout is a two-fold comparison, so its interval is wide by construction.

The eight-band decomposition assumes a 30 to 2000 Hz sensor, and the dataset does not provide one. Results are reported with the full eight bands to match the original pipeline, but three of them are constants.

5.5. Reproducibility

Every number in this paper is generated by a script from the released result files. The ladder, the feature builder, the figure generator and the macro generator are in `paper_q1`. The pre-registration, including the falsified prediction, is archived with the analysis code. Reporting follows the REFORMS recommendations for machine-learning-based science [27].

6. Conclusion

We reproduced the published row-level protocol on a public honeybee dataset, then removed one source of optimism at a time.

A random row split reaches 86.8 percent accuracy on the same features that reach 73.3 percent when the split respects hive membership.

At inspection granularity, hive leakage did not inflate the score. Gain-retaining features did not leak either. The only large effect at that unit of analysis came from indoor environment measurements, which added 8.5 to 11.3 macro-F1 points and transferred across apiaries.

Restoring the row-level protocol did inflate the score, by +9.2 points for a random forest and by +1.1 for logistic regression. Leakage therefore requires a row-level split and a model with enough capacity to memorise near-duplicate records.

The practical conclusion is narrow and testable. Aggregate before you split. Report the unit of analysis with the score. Do not attribute a high accuracy to the acoustic features until the split protocol is stated.

Data availability

The MSPB dataset is available at Zenodo record 11398835 under doi:10.1038/s41597-024-03695-1. Analysis code and result files are provided with this submission.

Declaration of competing interest

The authors declare no competing interests.

Appendix A. Pre-registered decision rules

The analysis plan, the predictions and the decision rules were written to disk before any model was fitted. This appendix reproduces the parts that constrain the reported conclusion.

Table A.3: Decision rules fixed before the analysis.

Outcome	Condition
Claim supported	The P0 to P2 gap exceeds twice the pooled bootstrap standard error, and P2 scores at least as high as P3.
Claim weakened	The gap lies between one and two standard errors. Report the result as inconclusive.
Claim falsified	The gap lies below one standard error. Report the apiary generalisation result alone.

Two further rules bind the analysis. The headline model is chosen at P2 before P3 is inspected. Every number in this paper is generated from a result file by a script.

Section 5 states that the claim is falsified. The decision rule was applied as written, and the result is reported without reframing.

Appendix B. Deviations from the plan

One deviation was made, before any model was fitted, and it is recorded here.

The original plan used the individual telemetry row as the unit of analysis, with a stress label derived from brood-nest temperature. Phase 1 profiling showed that this label was degenerate. 875,004 of 876,105 D2 rows fall below the 32 °C cut-off, which is 99.87 percent. The median D2 temperature is 18.3 degrees Celsius. A threshold that almost every row satisfies separates nothing.

The plan was revised to use the inspection event as the unit and colony

strength as the target. The row-level protocol was restored later as a control, not as the main analysis, and it is reported in Section 4.6.

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